

Chengze “Cheng” Liu

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MS CS (Data Analytics) | Python, SQL, Machine Learning | Seeking Data Science / ML Internship – Available June 2026

Education

University of Houston

Houston, TX

MS in Computer Science — Data Analytics Track

Jan 2026 – Present

BS in Computer Science | Minor in Mathematics GPA: 4.0

Aug 2023 – Dec 2025

Skills

Data & Analytics	Python, SQL, Pandas, NumPy, SciPy, Scikit-learn
Machine Learning	Regression, Classification, Clustering, Cross-Validation, Model Evaluation
Applied	NLP (TF-IDF, N-grams, POS Tagging), Deep Learning (PyTorch, TensorFlow), Geospatial (Shapely, Ripley's L)
Visualization	Matplotlib, Seaborn, Jupyter Notebook
Cloud & Tools	AWS (EC2, S3), Linux, GitHub, VS Code
Languages	English (Fluent), Chinese (Fluent)

Projects

Deep Learning Framework Benchmarking (Cloud Computing, AWS)

Mar 2026 – Apr 2026

- Designed a reproducible benchmarking study on AWS EC2 (g4dn.xlarge, NVIDIA T4) comparing PyTorch and TensorFlow CNN training across 12 controlled experimental runs, using CloudWatch and a custom GPU monitoring script for hardware-level metrics.
- Found PyTorch reached 80% CIFAR-10 validation accuracy $2.6\times$ faster (189s vs 494s) and at $2.6\times$ lower AWS cost (\$0.028 vs \$0.072), consuming $3\times$ less energy (2.4 vs 7.3 Wh).
- Traced the performance gap to execution architecture: TensorFlow's `@tf.function` graph compilation added GPU overhead that offset its efficient CPU data pipeline, while PyTorch's eager execution was faster overall despite higher CPU utilization (93% vs 45%).

Gender Prediction from Blog Posts (NLP, Machine Learning)

Mar 2026 – Apr 2026

- Achieved 72.5% gender classification accuracy (vs. 70% baseline) on 3,226 blog posts by systematically comparing linear SVM, logistic regression, and decision tree models with TF-IDF n-gram features.
- Diagnosed performance saturation through controlled ablation experiments across word-, character-, and POS-tag features; found stylometric signals degraded accuracy, revealing topical content as the dominant gender predictor.

Weather Data Outlier Detection (Data Mining, Unsupervised Learning)

Jan 2026 – Feb 2026

- Implemented and compared three unsupervised outlier detection methods — GMM, KDE, and a TensorFlow autoencoder — across 9 hyperparameter configurations on 365-day Houston weather data (6 features), validating each on a synthetic dataset with known outlier structure.
- Found that GMM uniquely captured rare multivariate anomalies (e.g., unusual wind-humidity co-occurrences), while the autoencoder concentrated on extreme rainfall due to its lightweight bottleneck architecture, and KDE reduced to univariate extremes detectable by a simple box plot — informing a concrete model selection recommendation.

Spatial Collocation Analysis of Urban Building Types (Data Mining, Spatial Statistics)

Mar 2026 – Apr 2026

- Parsed a 3,371-building geospatial KML dataset across two metadata encodings, projected coordinates from WGS84 to a metric grid (Lambert-93), and implemented Ripley's L-function and bivariate cross-L using `scipy cKDTree` for efficient radius queries.
- Designed a label-swap Monte Carlo null model (199 simulations, $\sim 34,000$ swaps each) to test collocation significance; confirmed commercial buildings cluster at all scales 5–500 m and showed raw nearest-neighbor distances incorrectly suggested proximity where the formal test detected anti-collocation.

Work Experience

Server / Front-of-House Staff

Sep 2021 – May 2022

Haidilao Hotpot | Katy, TX

- Managed order accuracy and customer flow in a high-volume restaurant; developed attention to detail and communication skills in a fast-paced team environment.